# ABHISHEK KUMAR

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## SUMMARY

B.Tech Computer Science (Artificial Intelligence) undergraduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Python-based application development. Experienced in building end-to-end AI projects involving real-time object detection, fire and smoke detection, fraud detection, model evaluation, and web-based AI applications. Interested in developing practical AI/ML solutions and pursuing opportunities as an AI Engineer, Machine Learning Engineer, or Software Engineer.

## TECHNICAL SKILLS

**Programming:** Python, JavaScript, SQL

**Machine Learning:** Scikit-learn, Pandas, NumPy, Matplotlib, Classification, Model Evaluation

**AI & Computer Vision:** YOLO, OpenCV, Deep Learning, Object Detection

**Backend & Web:** Flask, Streamlit, HTML, CSS

**Databases:** MySQL, SQLite

**Tools:** Git, GitHub, VS Code, Power BI, Microsoft Excel

## PROJECTS

**SentinelAI — AI-Powered Fire, Smoke & Intrusion Detection System**

*Python | YOLO | OpenCV | Flask | SQLAlchemy*

- Developed a real-time computer vision system for detecting fire and smoke from camera input using a custom YOLO model.
- Implemented automated alerts, evidence storage, database integration, and a web-based monitoring dashboard.
- Integrated YOLO object detection with a Flask backend for real-time detection, alert management, and evidence retrieval.
- Designed the system for practical emergency monitoring and real-time visual event detection.

**Credit Card Fraud Detection**

*Python | Machine Learning | Scikit-learn | Pandas*

- Built a machine-learning classification workflow to identify potentially fraudulent credit card transactions.
- Performed data preprocessing, train-test splitting, model evaluation, and probability-threshold tuning.
- Focused on improving fraud detection recall while evaluating classification performance using appropriate metrics.

**Real-Time Object Detection**

*Python | YOLO | OpenCV*

- Developed a live camera-based object detection application using YOLO and OpenCV.
- Integrated a YOLO detection model with webcam input for real-time visual recognition.
- Configured and tested the application in a local Python development environment.

**Train Booking System**

*Python | SQL | Web Development*

- Developed a web-based railway booking application for managing train schedules, passenger information, seat availability, andbookings.
- Implemented application workflows around practical reservation and database-management requirements.
- Applied Python and SQL concepts to build a functional data-driven application.

## EDUCATION

**Bansal Institute of Engineering and Technology, AKTU**

**B.Tech in Computer Science and Engineering (Artificial Intelligence)** | 2023 – 2027

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, DBMS, Machine Learning, Computer Vision

**R.P.P.S. Inter College, Ambedkar Nagar** — Intermediate (12th)

**R.P.P.S. Inter College, Ambedkar Nagar** — High School (10th)

## TRAINING & CERTIFICATIONS

- Python and AI-focused practical training with hands-on programming and project-based development.
- Project-based experience in Machine Learning, Computer Vision, Python, and AI application development.

## ADDITIONAL INFORMATION

**Career Interests:** AI Engineer, Machine Learning Engineer, Software Engineer

**Soft Skills:** Problem-Solving, Communication, Teamwork, Quick Learning, Time Management